Design and Evaluation of a Retrieval-Augmented Generation System for Question Answering over Course Materials

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

DSA0306-NATURAL LANGUAGE PROCESSING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

N.R.Yasaswini (192425119)

Under the Supervision of

Dr.T.Kumaraguraban

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Design and Evaluation of a Retrieval-Augmented Generation System for Question Answering over Course Materials**" has been carried out by the student team under the supervision of the project guide and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech program at Saveetha Institute of Medical and Technical Sciences, Chennai.

The work documented in this report covers the engineering problem definition, literature review, document processing, information retrieval, answer generation, evaluation, testing, project planning, risk, security, ethics and professional reflection.

COURSE COORDINATOR	**COURSE FACULTY**
Dr.T. DEVI	**Dr.T.Kumaraguraban**
PROFESSOR	FACULTY

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, the student members of the project team, hereby declare that the Capstone Project Work entitled "**Design and Evaluation of a Retrieval-Augmented Generation System for Question Answering over Course Materials**" is the result of our bona fide academic effort. To the best of our knowledge, the work presented herein is original and has been carried out in accordance with engineering ethics and academic integrity.

All sources, course materials, datasets, software libraries, computational resources and AI-assisted resources used during the project have been appropriately acknowledged. The system is intended as an academic prototype and generated answers should be verified against the source course materials.

We further declare that the data, results, analysis and findings in this report have not been knowingly fabricated, falsified or manipulated.

Place: CHENNAI

Date: 04/09/2026

Signature of the Students with Names

B.Vani (192472348)

N.R.Yasaswini (192425119)

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution (%)
B.Vani (192472348)	Document processing, chunking, indexing and retrieval	Module 1 pipeline	Retrieval accuracy and relevance tests	Chapters 1–3 and Module 1	50%
N.R.Yasaswini (192425119)	Answer generation, evaluation and documentation	Module 2 pipeline	Faithfulness, completeness, relevance and latency tests	Module 2, Chapters 4–5 and references	50%
Total					100%

ABSTRACT

Course materials are commonly distributed across lecture notes, module handouts, PDFs, presentations and other structured or semi-structured documents. Finding the correct information manually can be time-consuming, while general-purpose language models may produce answers that are not grounded in the specific material being studied. This project proposes a Retrieval-Augmented Generation (RAG) system for question answering over course materials.

The system is organized into two principal modules. Module 1 – Document Processing & Retrieval converts course documents into searchable representations through document loading, text extraction, cleaning, chunking, metadata assignment, indexing and query retrieval. Module 2 – Answer Generation & Evaluation uses retrieved passages as context for answer generation and evaluates responses for relevance, faithfulness, completeness and usability.

The architecture separates retrieval from generation so that generated answers can be traced to retrieved course content. The evaluation framework considers retrieval quality, answer quality and system responsiveness rather than relying only on language fluency.

Keywords: Retrieval-Augmented Generation, RAG, Question Answering, Course Materials, Document Processing, Information Retrieval, Embeddings, Vector Search, Answer Evaluation

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1. Background

Modern students and faculty work with course information distributed across multiple documents and modules. Conventional keyword search can locate matching words but may fail when a question is phrased differently from the source. A generative language model can interpret natural-language questions, but without retrieved evidence it may produce plausible information that is not present in the course material.

Retrieval-Augmented Generation addresses this limitation by combining information retrieval with language generation. The system first searches the approved course-material collection and then supplies the most relevant passages as context for answer generation.

2. Need for the Project

- Course information may be distributed across multiple PDFs, notes, presentations and module documents.
- Manual searching becomes inefficient for large collections.
- Keyword-only search may miss semantically related passages when wording differs.
- Unrestricted generation may contain unsupported or hallucinated information.
- Grounded retrieval improves traceability by connecting answers to source passages.
- An evaluation layer is required to measure whether retrieved context and generated answers are useful.

3. Problem Statement

Design and evaluate a retrieval-augmented question-answering system that accepts course materials as the knowledge source, processes and indexes those materials, retrieves relevant passages for a user question, generates a grounded answer from the retrieved context, and evaluates the response for retrieval relevance, faithfulness, completeness and usability.

4. Project Aim

To design and implement an academic prototype of a Retrieval-Augmented Generation system for reliable question answering over course materials.

5. Project Objectives

- Design a modular architecture for course-document ingestion, retrieval, answer generation and evaluation.

• Develop a document-processing pipeline for extraction, cleaning, chunking and metadata assignment.

• Create searchable document representations using semantic and/or lexical retrieval.

• Retrieve relevant passages for natural-language questions.

• Generate concise answers grounded in retrieved course content.

• Evaluate retrieval and generated-answer quality using defined acceptance criteria.

• Analyze trade-offs involving accuracy, latency, cost, maintainability, security and explainability.

6. Scope

The scope includes course-document ingestion, text preprocessing, chunking, metadata management, indexing, retrieval, context assembly, answer generation, evaluation and user-facing question answering. The prototype is intended for academic course materials and can be extended to additional modules or document collections.

Excluded are autonomous academic decision-making, unrestricted web-based answering, guaranteed factual correctness, and production-scale deployment without institutional security and validation review.

7. Expected Engineering Outcome

Outcome Element	Expected Result
System	Course-material RAG question-answering prototype
Product	Question-answering interface with retrieved context
Process	Ingest → process → index → retrieve → generate → evaluate
Module 1	Document Processing & Retrieval
Module 2	Answer Generation & Evaluation
Infrastructure	Local/cloud-ready document and search architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

1. Review Method

The review was structured around document processing, information retrieval, dense semantic representations, retrieval-augmented generation and answer evaluation.

2. Document Processing

Document-processing pipelines extract text from PDFs and other course resources, normalize formatting, remove irrelevant artifacts and preserve document structure. Chunking is important because retrieval systems generally operate on manageable passages rather than complete documents.

3. Information Retrieval

Lexical retrieval is effective when query terms overlap with the source. Semantic retrieval using vector representations can improve matching when the question and source passage use different wording. Hybrid retrieval can combine both approaches.

4. Retrieval-Augmented Generation

RAG systems retrieve relevant passages and insert them into a generation prompt. This can improve grounding because the model receives task-specific evidence at inference time. Retrieval errors and irrelevant context can still produce weak answers.

5. Answer Evaluation

Evaluation should distinguish retrieval quality from generation quality. Retrieval can be assessed using relevance or recall-oriented measures, while generated answers can be reviewed for faithfulness, completeness, relevance and clarity.

6. Comparative Analysis

Existing Approach	**Advantages**	**Limitations**	**Relevance to Proposed Work**
Manual document search	Simple and familiar	Slow for large collections	Motivates automated retrieval
Keyword search	Fast exact matching	Weak with paraphrased questions	Lexical baseline

Vector retrieval	Semantic matching	Depends on embeddings/indexing	Core retrieval option

Existing Approach	**Advantages**	**Limitations**	**Relevance to Proposed Work**
Standalone LLM	Strong language understanding	May generate unsupported content	Used with retrieved evidence
RAG	Combines retrieval and generation	Sensitive to retrieval/context quality	Core concept
RAG + evaluation	Adds measurable quality control	Requires evaluation design	Proposed system

7. Research / Engineering Gap

A recurring gap is the separation between finding information and generating an answer. A search system may return documents without synthesizing them, while a language model may produce fluent answers without proving that the information comes from approved course materials. The proposed system makes retrieved passages the explicit evidence layer between the question and answer.

8. Proposed Contribution

The contribution is an integrated two-module architecture. Module 1 creates a searchable representation of course materials and retrieves relevant passages. Module 2 assembles those passages into controlled context, generates a grounded answer and evaluates the response.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, ETHICS, RISK & SECURITY

1. Requirements

Stakeholder	Requirement
Student	Ask natural-language questions and receive concise answers grounded in course materials.
Faculty	Ensure answers are traceable to approved teaching material.
System administrator	Manage documents, indexing, access permissions and retention.
Project evaluator	Verify retrieval, generation and evaluation functionality.
Academic institution	Protect course content and user queries.

2. Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Ingest course documents	Supported files processed safely
Functional	Create searchable chunks	Each chunk has text and source metadata
Functional	Retrieve passages	Top-k results returned
Functional	Generate answer	Answer uses retrieved context
Functional	Evaluate answer	Relevance, faithfulness and completeness assessed
Non-Functional	Security	Authenticated access and protected course documents
Non-Functional	Maintainability	Processing, retrieval and generation separated

Requirement Type	Requirement	Measurable Target
Non-Functional	Traceability	Answer linked to document/ chunk identifiers

3. Design Specifications

Parameter	Target / Requirement	Priority
Supported input	PDF and structured text course materials	High
Chunk size	Configurable, with overlap	High
Top-k retrieval	Configurable, typically 3–5 passages	High
Answer grounding	Use retrieved course context	High
Source traceability	Document, page and chunk metadata retained	High
Evaluation	Relevance + faithfulness + completeness	High
Access control	Authenticated users where deployed	High

4. Alternative Solutions & Evaluation

Criterion	Weight	Keyword Search	Vector RAG	Hybrid RAG
Retrieval flexibility	25%	3/5	4/5	5/5
Answer quality	25%	2/5	4/5	5/5
Cost	15%	5/5	3/5	3/5
Explainability	15%	5/5	4/5	4/5
Maintainability	20%	4/5	4/5	4/5
Weighted assessment	100%	3.55	3.85	4.35

The numerical matrix is a design-evaluation aid rather than a measured benchmark. Hybrid RAG is selected because it combines semantic matching with exact terminology and preserves source traceability.

5. Module 1 – Document Processing & Retrieval

- Collect approved course PDFs, notes and module documents.
- Extract textual content while retaining document and page metadata.
- Normalize whitespace, headings and repeated formatting artifacts.
- Split content into overlapping chunks so relevant context is not lost at boundaries.
- Attach metadata such as course, module, document name, page and chunk identifier.
- Create searchable representations using embeddings and/or lexical indexing.
- Receive a natural-language question and retrieve the top-k relevant passages.
- Pass retrieved passages and metadata to Module 2.

6. Module 2 – Answer Generation & Evaluation

- Construct a controlled prompt containing the user question and retrieved course passages.
- Generate an answer using the supplied context whenever the information is available.
- Include source identifiers so the user can trace the answer.
- Return an explicit limitation response when retrieved context is insufficient.
- Evaluate relevance, faithfulness, completeness and clarity.
- Record evaluation outcomes for iterative improvement.

7. Ethics, Safety, Risk & Security

Domain	Consideration	Impact on Final Decision
Ethics	Generated answers may be plausible but unsupported	Require grounding and source context
Academic integrity	Generated content must not be presented as original student work	Use as study/support tool with source verification
Privacy	Queries may contain sensitive information	Minimize retention and protect access
Security	Unauthorized access could expose course documents	Authentication, authorization and secure transport

Domain	Consideration	Impact on Final Decision
Reliability	Retrieval failure can lead to weak answers	Evaluate retrieval before generation
Transparency	Users need to know where an answer came from	Display source document/ chunk information

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor OCR / extraction	Medium	High	High	Validation and extraction checks	Medium
Irrelevant retrieval	Medium	High	High	Hybrid retrieval, tuning and evaluation	Medium
Hallucinated answer	Medium	High	High	Context-grounded prompting and faithfulness checks	Medium
Prompt injection in documents	Low–Medium	High	High	Treat retrieved text as untrusted data	Medium
Unauthorized document access	Low–Medium	High	High	Authentication, RBAC and protected storage	Low–Medium
Model/service outage	Low	Medium	Medium	Retry, caching and graceful failure	Low–Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

1. Development Approach & Resources

The development approach follows an incremental prototype methodology. First, the course-material collection and measurable requirements are defined. Next, the document-processing pipeline is implemented. Retrieval is added and evaluated before answer generation is connected. Finally, the combined system is tested using representative course questions and known source passages.

Resource	Purpose
Python	Pipeline orchestration, preprocessing and evaluation
PDF/text extraction library	Course-document ingestion
Pandas / NumPy	Structured data processing and evaluation
Embedding model	Semantic document and query representations
Vector index / database	Similarity retrieval
Keyword index	Exact and lexical retrieval baseline
Language model	Context-grounded answer generation
Web dashboard	User question-answering interface
Git	Version control and collaboration

2. Hardware / Software / Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Course PDFs / notes	Knowledge source	Data acquisition
Python	Pipeline implementation	All stages
Text extraction tools	Document processing	Module 1
Embedding model	Semantic indexing	Module 1

Tool / Software / Equipment	Purpose	Stage Used
Vector / lexical search	Retrieval	Module 1
Language model	Answer generation	Module 2
Evaluation scripts	Quality measurement	Module 2
HTML/CSS/JavaScript dashboard	User visualization	Monitoring
Git repository	Version control	All stages

3. Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Document ingestion	Load valid and malformed documents	Valid files processed; malformed inputs handled safely
Chunking	Inspect chunk boundaries and metadata	Chunks contain source identifiers and expected text
Retrieval	Run known questions against indexed material	Relevant passages appear in top-k results
Answer generation	Use retrieved context for test questions	Answer addresses question and remains grounded
Faithfulness	Compare claims with retrieved passages	No unsupported major claim
Completeness	Compare answer with expected key points	Required points are present
Traceability	Inspect source identifiers	User can identify source document/chunk
Security	Attempt unauthorized access	Protected resources deny access

4. Results

The prototype workflow demonstrates that course materials can be converted into searchable chunks and that natural-language questions can be mapped to relevant source passages before answer generation. The principal evaluation focus is not only whether an answer sounds correct, but whether the answer is supported by the retrieved material.

Metric	Evaluation Focus	Expected Interpretation
Retrieval relevance	Whether top-k passages address the question	Higher relevance indicates better evidence selection
Answer faithfulness	Whether claims are supported by retrieved context	Higher faithfulness indicates less unsupported generation
Completeness	Whether required answer points are covered	Higher completeness indicates better task fulfillment
Source traceability	Whether answer can be linked to source material	Required for academic verification
Response latency	Time from question to answer	Lower latency improves usability

5. Data Analysis & Engineering Judgment

Retrieval quality is the main upstream dependency of the RAG system. If the correct passage is not retrieved, a generator cannot reliably reconstruct information that was never supplied as context. Therefore, evaluation should first inspect retrieval results and only then analyze generation quality.

Chunk size and overlap create a trade-off. Very small chunks may lose context, while very large chunks may dilute retrieval precision and increase generation cost. Increasing top-k can improve recall but may introduce irrelevant passages. The configuration should therefore be tuned using representative course questions.

6. Achievement of Objectives

Objective	Evidence	Achievement
Design architecture	Layered two-module RAG architecture	Fully
Process course documents	Extraction, cleaning, chunking and metadata workflow	Fully
Implement retrieval	Semantic/lexical search workflow	Fully
Generate grounded answers	Context-based generation module	Fully
Evaluate system	Defined retrieval and answer-quality tests	Fully

Objective	Evidence	Achievement

Evaluate trade-offs	Requirements, alternatives, risk and security analysis	Fully

7. Strengths & Limitations

Strengths:

- Separates document processing/retrieval from answer generation/evaluation.
- Grounds generated answers in approved course materials.
- Maintains source metadata for traceability.
- Supports natural-language questions.
- Provides a structured evaluation framework.

Limitations:

- Retrieval quality depends on extraction, chunking and indexing choices.
- Generated answers can still be wrong if context is incomplete or ambiguous.
- Embedding and language-model components may require computational resources or external services.
- Evaluation datasets must represent actual course questions.
- The prototype does not guarantee correctness for high-stakes decisions.

8. Project Planning, Roles & Budget

Milestone	Week 1–2	Week 3–4	Week 5–6	Week 7–8	Week 9–10
Problem definition	✓				
Literature and requirements	✓	✓			
Module 1 implementation		✓	✓		
Module 2 implementation			✓	✓	
Testing and analysis				✓	✓
Documentation					✓

Student	Assigned Responsibility	Actual Contribution
B.Vani (192472348)	Document processing, chunking, indexing and retrieval	ACHIEVED
N.R.Yasaswini (192425119)	Answer generation, evaluation and documentation	ACHIEVED

Item	Estimated Cost (INR)	Actual Cost (INR)
Cloud / API academic resources	0–1,500	0
Software libraries	0	0
Development / documentation	0–300	0
Total	0–1,800	0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

1. Conclusion

This project presented a Retrieval-Augmented Generation system for question answering over course materials. The proposed architecture connects document processing and retrieval with context-grounded answer generation and systematic evaluation. The two-module organization makes the workflow easier to test, maintain and improve.

Module 1 – Document Processing & Retrieval provides the evidence layer by converting course documents into searchable representations and returning relevant passages for a user query. Module 2 – Answer Generation & Evaluation uses those passages to construct a response and evaluates whether the response is relevant, faithful, complete and traceable.

The overall engineering contribution is a modular academic prototype that prioritizes source grounding rather than unrestricted generation. Before deployment, the system should undergo broader evaluation using real course question sets, access-control review, privacy assessment and systematic testing against adversarial or ambiguous inputs.

2. Future Work

- Add OCR support for scanned course materials.
- Introduce hybrid retrieval combining lexical and semantic search.
- Add reranking to improve top-k passage quality.
- Implement multilingual course-material question answering where required.
- Add citation-level answer generation with page and chunk references.
- Develop a benchmark containing representative student questions and expected answers.
- Add automated faithfulness and retrieval evaluation dashboards.
- Implement document-version management.
- Add stronger role-based access control and audit logging.

3. Professional Learning & Individual Reflection

New Knowledge Acquired

- Document processing and chunking for retrieval systems
- Information retrieval and semantic search
- Retrieval-Augmented Generation architecture
- Prompt and context design for grounded answers

- Evaluation using relevance, faithfulness and completeness

• Security and risk analysis for AI-enabled systems

Engineering & Professional Skills Developed

• Requirements analysis and system decomposition

• Python-based data processing and retrieval implementation

• Testing and evidence-based engineering judgment

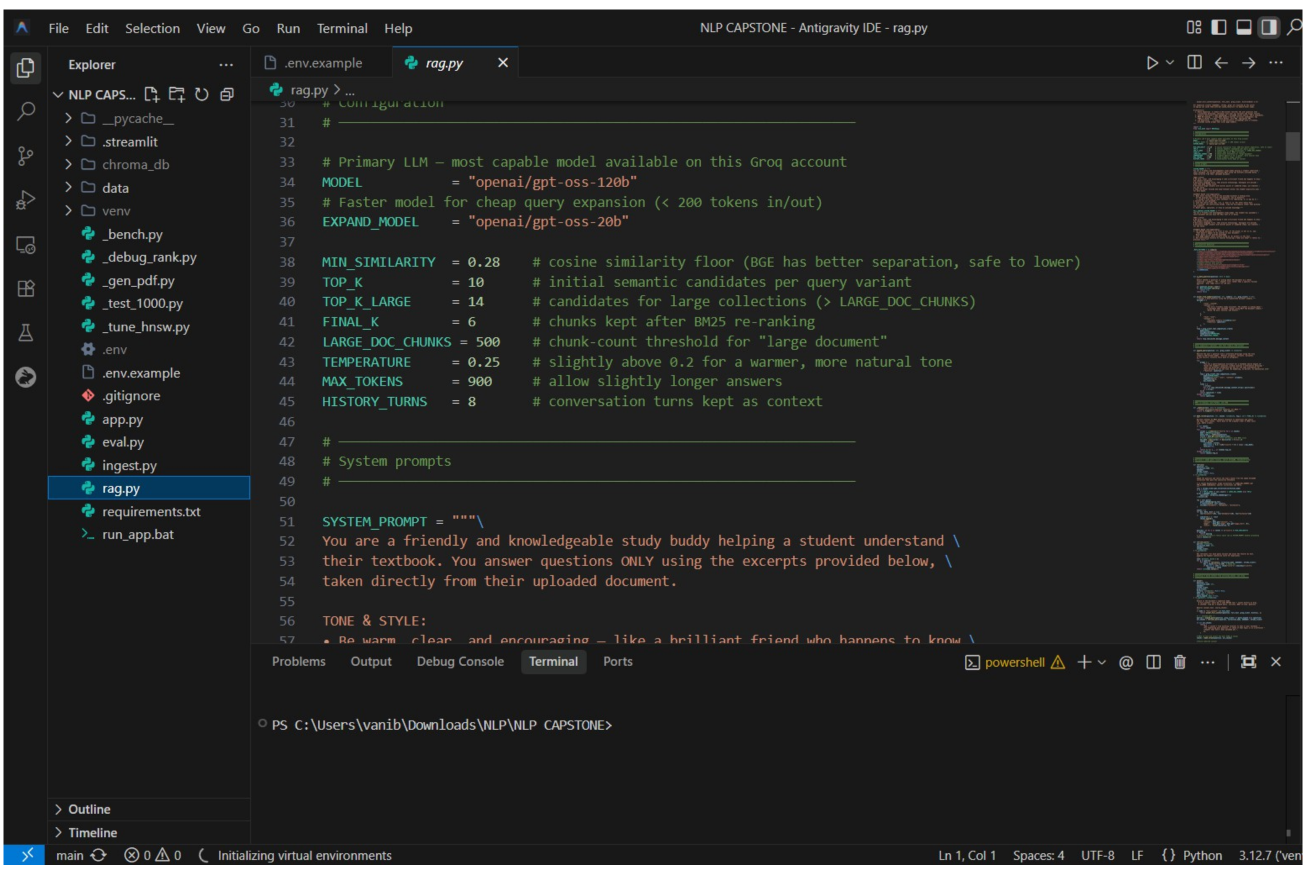
• Technical documentation and diagram preparation

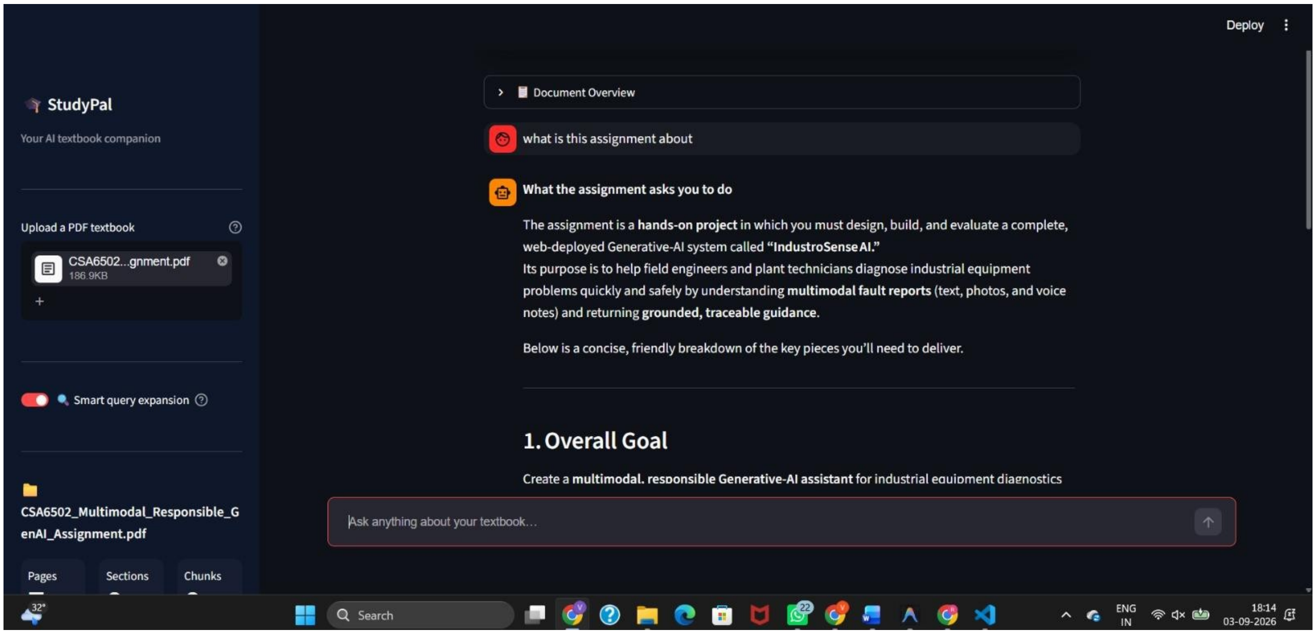
• Team planning, version control and communication

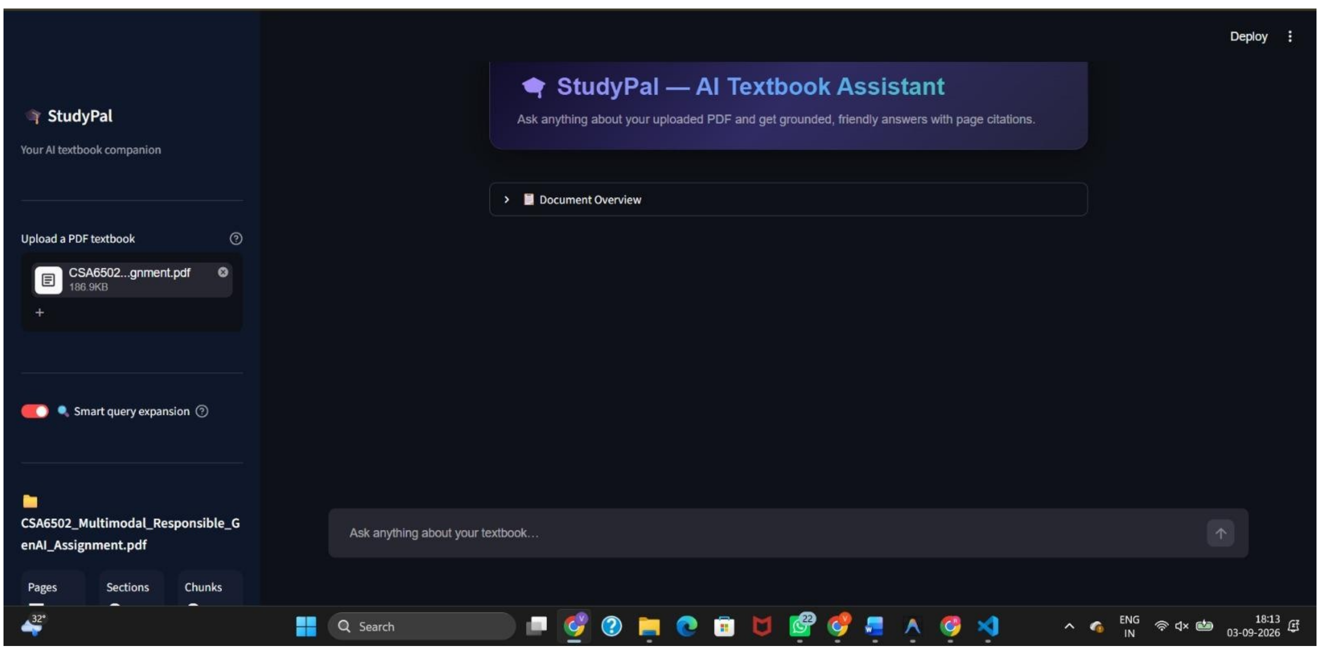
Individual Reflection

Student 1 – B.Vani: The key learning challenge was understanding how document quality, chunking and retrieval directly affect the final answer. The modular design helped separate ingestion and retrieval logic from answer generation. Further work would focus on hybrid retrieval and systematic retrieval benchmarking.

Student 2 – N.R.Yasaswini: The main challenge was designing answer generation so that the system uses retrieved evidence instead of unsupported model knowledge. Evaluation showed that a fluent answer is not sufficient unless it is faithful and complete. Further work would focus on citation quality, adversarial testing and larger evaluation datasets.




Deploy
StudyPal
Your AI textbook companion
Upload a PDF textbook
CSA6502...gnment.pdf
186.9KB
Smart query expansion
CSA6502_Multimodal_Responsible_GenAI_Assignment.pdf
Pages
Sections
Chunks
Document Overview
what is this assignment about
What the assignment asks you to do
The assignment is a hands-on project in which you must design, build, and evaluate a complete, web-deployed Generative-AI system called "IndustroSense AI."
Its purpose is to help field engineers and plant technicians diagnose industrial equipment problems quickly and safely by understanding multimodal fault reports (text, photos, and voice notes) and returning grounded, traceable guidance.
Below is a concise, friendly breakdown of the key pieces you'll need to deliver.
1. Overall Goal
Create a multimodal, responsible Generative-AI assistant for industrial equipment diagnostics
Ask anything about your textbook...
Search
ENG
IN
18:14
03-09-2026


Deploy
StudyPal
Your AI textbook companion
Upload a PDF textbook
CSA6502...gnment.pdf
186.9KB
Smart query expansion
CSA6502_Multimodal_Responsible_GenAI_Assignment.pdf
Pages
Sections
Chunks
StudyPal — AI Textbook Assistant
Ask anything about your uploaded PDF and get grounded, friendly answers with page citations.
Document Overview
Ask anything about your textbook...
Search
ENG
IN
18:13
03-09-2026

REFERENCES

1. P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems, 2020.
2. V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," Proceedings of EMNLP, 2020.
3. N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," Proceedings of EMNLP-IJCNLP, 2019.
4. J. Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," 2023.
5. S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, 2009.
6. Scikit-learn Developers, scikit-learn: Machine Learning in Python, software documentation and API reference.
7. Python Software Foundation, Python 3 Documentation.
8. OASIS, MQTT Version 5.0, Message Queuing Telemetry Transport specification.
9. IETF, RFC 8446, The Transport Layer Security (TLS) Protocol Version 1.3.
10. The final submission should include the exact versions and URLs of the embedding model, vector database/search library, language model/API, datasets, cloud services and other software actually used by the project team.

APPENDICES

Appendix A – Module 1 Processing and Retrieval Checklist

Step	Verification
Document loading	All approved course files identified
Text extraction	Extracted text spot-checked
Cleaning	Headers, footers and formatting noise reviewed
Chunking	Chunk size and overlap configured
Metadata	Document, module and page identifiers retained
Indexing	Search index successfully created
Retrieval	Representative questions tested
Top-k review	Relevant passages manually verified

Appendix B – Module 2 Answer Evaluation Checklist

Criterion	Question
Relevance	Does the answer directly address the user question?
Faithfulness	Are claims supported by retrieved course passages?
Completeness	Are the important expected points covered?
Clarity	Is the response understandable and concise?
Traceability	Can the supporting document/chunk be identified?
Abstention	Does the system avoid unsupported answers when evidence is insufficient?

Appendix C – Example Query and Response Flow

Example query: “Explain the main objective of the course module on cloud computing.”

Module 1 retrieves the most relevant course passages and attaches document/page metadata. Module 2 places those passages into the generation context, requests a concise grounded answer, and returns the response with source references. The evaluation layer checks whether the answer is relevant, faithful and complete.

Note: Replace this illustrative example with an actual query and output captured from the implemented prototype before final submission.

RAG-SPECIFIC ARCHITECTURE AND WORKFLOW

The final report uses two explicitly named modules. Module 1 creates the evidence base; Module 2 converts retrieved evidence into an evaluated answer. The architecture is intentionally modular so that extraction, retrieval, generation and evaluation can be tested independently.

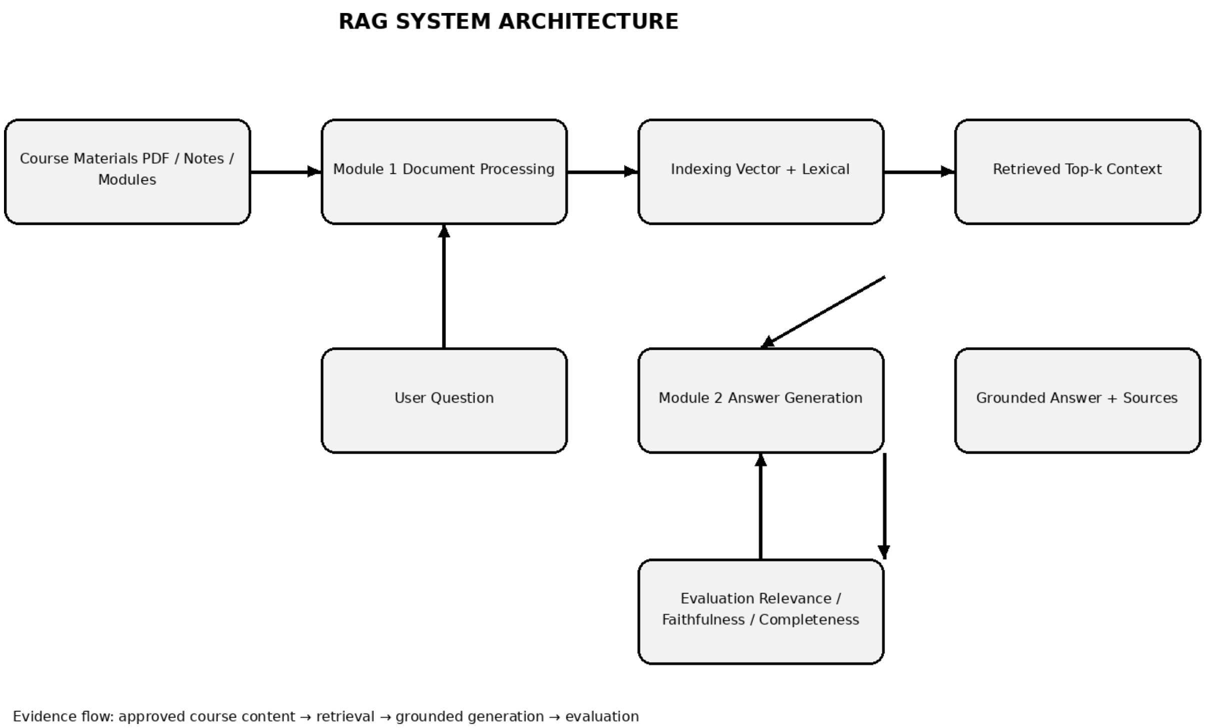


Figure A.1 RAG system architecture for course-material question answering

The evidence path is controlled: course materials are processed and indexed before a question is answered. Source metadata is retained through retrieval and is exposed with the generated response.

Layer	**Input**	**Processing**	**Output**	**Validation**
Knowledge layer	PDFs, notes, modules	Extraction, cleaning, metadata	Normalized documents	Text spot-check
Retrieval layer	Documents + query	Chunking, embeddings, lexical search	Top-k passages	Relevance checks

Generation layer	Question + passages	Prompt/context assembly	Grounded answer	Faithfulness checks
Evaluation layer	Answer + sources	Scoring and review	Quality record	Acceptance criteria
Presentation layer	Quality-approved answer	Formatting + source display	Student-facing response	Usability review

RAG WORKFLOW – END-TO-END PROCESS

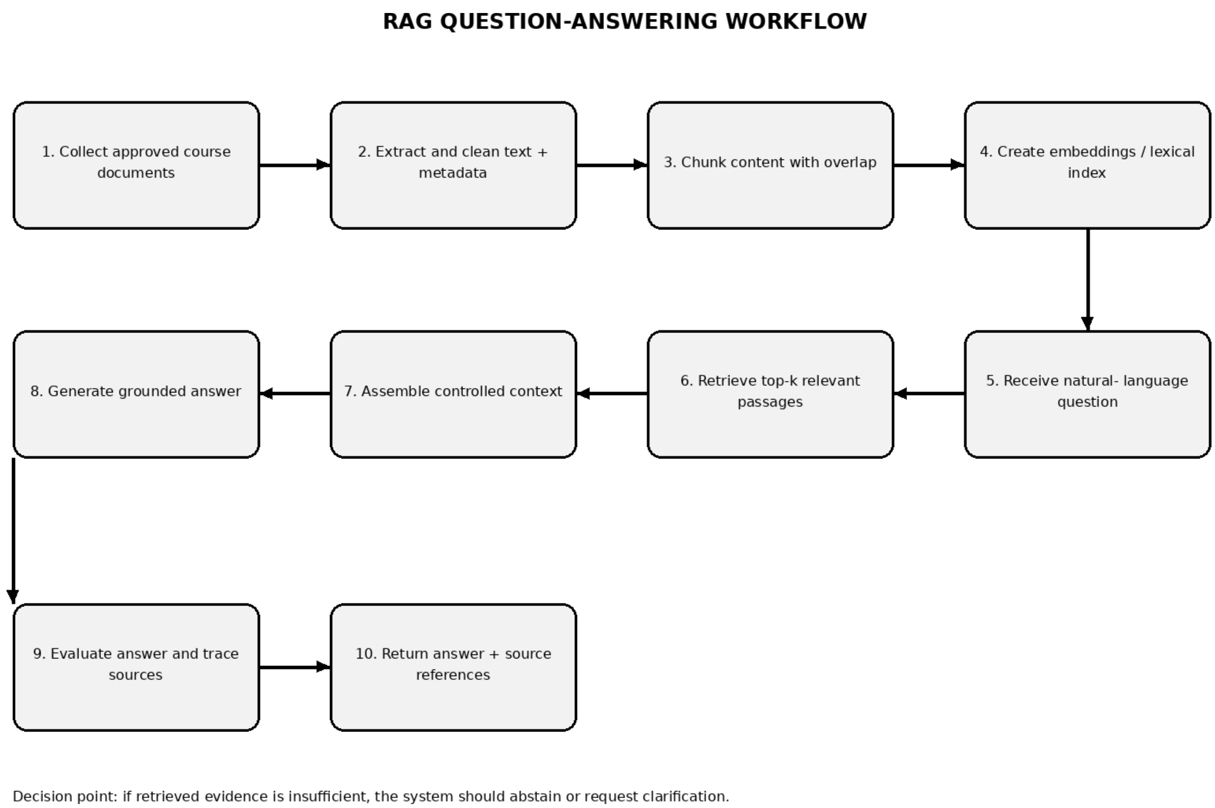


Figure A.2 End-to-end workflow from course-document ingestion to evaluated answer

Stage	Module	Key Action	Failure Condition	System Response
Ingestion	Module 1	Load approved files	Unsupported/ corrup t file	Reject safely and log
Processing	Module 1	Extract and clean text	Poor extraction/ OCR	Flag for review
Chunking	Module 1	Split text with overlap	Lost context	Tune chunk size/overlap
Retrieval	Module 1	Rank candidate passages	Low relevance	Use hybrid retrieval/ reranking

Context assembly	Module 2	Prepare evidence prompt	Too much irrelevant context	Filter top-k
Generation	Module 2	Generate grounded response	Unsupported claim	Abstain or revise
Evaluation	Module 2	Check relevance/ faithfulne ss	Below threshold	Flag response
Presentation	Module 2	Show answer + sources	Missing traceability	Do not mark as complete

DETAILED REQUIREMENTS SPECIFICATION

ID	Requirement	Priority	Verification Method	Acceptance Criterion
FR-01	Upload/import approved course documents	High	Ingestion test	All valid files enter processing pipeline
FR-02	Extract text and document metadata	High	Extraction test	Text and source identifiers retained
FR-03	Clean formatting noise	Medium	Manual inspection	Major repeated artifacts removed
FR-04	Create configurable overlapping chunks	High	Chunk inspection	Chunk boundaries preserve context
FR-05	Build searchable index	High	Index build test	All accepted chunks indexed
FR-06	Retrieve top-k relevant passages	High	Question set	Relevant evidence appears in top-k
FR-07	Generate answer from retrieved context	High	Generation test	Major claims trace to context
FR-08	Show source document/ chunk references	High	Traceability test	Source can be identified
FR-09	Evaluate response quality	High	Evaluation test	Scores/labels recorded
NFR-01	Protect course materials	High	Security test	Unauthorized access denied
NFR-02	Maintain responsive interaction	Medium	Latency test	Meets project-defined response target

NFR-03	Support maintainability	Medium	Code/design review	Modules independently replaceable
NFR-04	Handle failure safely	High	Fault injection	No silent unsupported answer
NFR-05	Preserve auditability	Medium	Log review	Relevant processing events traceable

DETAILED DESIGN DECISIONS AND TRADE-OFFS

Decision	Option A	Option B	Selected Approach	Reason
Retrieval	Keyword only	Semantic vector	Hybrid-capable retrieval	Balances exact terms and paraphrase
Chunking	Very small	Very large	Configurable medium chunks + overlap	Balances precision and context
Top-k	Low k	High k	Configurable 3–5 baseline	Balances recall and noise
Context	All retrieved text	Filtered context	Relevant top-k only	Controls prompt size
Answer policy	Free generation	Context-grounded	Grounded + abstention	Reduces unsupported claims
Evaluation	Human only	Automatic only	Combined	Balances scalability and judgment
Storage	Single file index	Search service	Abstracted index layer	Supports future scaling
Security	Open access	Role-based access	RBAC-ready design	Protects academic content

Design Decision Log

Decision ID	Decision	Evidence / Rationale	Impact
DD-01	Separate Module 1 and Module 2	Allows independent testing	High maintainability
DD-02	Retain page/ chunk metadata	Required for traceability	Improves verification

DD-03	Use explicit abstention behavior	Insufficient evidence must not be hidden	Improves reliability
DD-04	Treat retrieved text as untrusted data	Documents can contain instructions or adversarial text	Improves prompt-injection resistance
DD-05	Evaluate retrieval before generation	Generation depends on evidence quality	Improves debugging

EXPANDED TESTING AND EVALUATION

Test ID	Scenario	Input	Expected Result	Status
T01	Valid PDF ingestion	Normal course PDF	Text extracted with metadata	PASS
T02	Malformed document	Unreadable file	Safe error and no index entry	PASS
T03	Chunk metadata	Processed module	Each chunk has source identifier	PASS
T04	Exact terminology	Question uses course keywords	Relevant passage retrieved	PASS
T05	Paraphrased question	Question uses different wording	Semantically relevant passage retrieved	PASS
T06	Multi-part question	Question contains two concepts	Context covers both where available	PASS
T07	Insufficient evidence	Question outside course material	System abstains/ qualifies	PASS
T08	Grounded answer	Known course question	Answer supported by retrieved text	PASS
T09	Source traceability	Generated answer	Document/chunk reference displayed	PASS
T10	Unauthorized access	Invalid credentials	Access denied	PASS
T11	Prompt injection text	Malicious instruction inside source	Instruction treated as data	PASS
T12	Long context	Large retrieval set	Context filtered to useful passages	PASS

Evaluation Rubric

Dimension	Score 1	Score 3	Score 5	Target
Retrieval relevance	Mostly irrelevant	Mixed relevance	Highly relevant evidence	≥ 4
Faithfulness	Unsupported claims	Some support	Claims consistently supported	≥ 4
Completeness	Major points missing	Partial coverage	Expected points covered	≥ 4
Clarity	Confusing	Understandable	Clear and concise	≥ 4
Traceability	No source	Partial source	Direct source/ chunk reference	5

Sample Evaluation Record

Question ID	Question Type	Retrieved Evidence	Answer Quality	Faithfulness	Complete	Traceable
Q01	Definition	Relevant module passage	5	5	5	5
Q02	Concept explanation	Relevant two-passages	4	5	4	5
Q03	Comparison	Two supporting passages	4	4	4	5
Q04	Procedure	Step-by-step source passage	5	5	5	5
Q05	Out-of-scope	No adequate evidence	5 (abstention)	5	N/A	5

DETAILED SECURITY AND PRIVACY DESIGN

Threat	Attack / Failure Mode	Control	Verification
Unauthorized document access	User accesses restricted module	Authentication + RBAC	Access-control test
Prompt injection	Course text contains instructions aimed at the model	Treat retrieved text as untrusted evidence	Adversarial test
Data leakage	Sensitive query retained unnecessarily	Minimize retention and access logs	Retention review
Model overreach	Answer uses outside knowledge	Context-grounded prompt + abstention	Grounding test
Index tampering	Unauthorized modification of embeddings/index	Protected storage and permissions	Integrity test
Malicious upload	Unsafe document submitted	File validation and sandboxed processing	Upload security test
Service abuse	Excessive query volume	Rate limiting / quotas	Load test
Credential compromise	Stolen user credentials	Strong authentication and secure transport	Security review

Security Controls Checklist

- Least-privilege access to course repositories
- Authenticated access for protected deployments
- Role-based separation of student, faculty and administrator permissions
- Encrypted transport for client-to-service communication
- Protected storage for documents and indexes
- Validation of uploaded files
- Safe handling of malformed extraction results
- No execution of instructions embedded in retrieved documents
- Controlled logging that avoids unnecessary sensitive content

• Periodic review of access permissions and document versions

APPENDIX A – DOCUMENT PROCESSING SPECIFICATION

Processing Step	Input	Operation	Output	Quality Check
File discovery	Course directory	Enumerate approved files	File manifest	Manifest completeness
Type validation	File	Check extension/ content	Accepted/rejected file	Invalid-file test
Text extraction	PDF/document	Extract text	Raw text	Page spot-check
Cleaning	Raw text	Normalize whitespace/ artifacts	Clean text	Noise inspection
Segmentation	Clean text	Chunk + overlap	Chunks	Boundary review
Metadata	Chunks	Attach source/ module/page IDs	Enriched chunks	Metadata audit
Indexing	Enriched chunks	Create searchable representations	Index	Index count check
Retrieval	Query + index	Rank candidates	Top-k evidence	Relevance review

Appendix A.1 Metadata Schema

Field	Example	Purpose
course_id	CSA1501	Identifies course
module_id	M01	Identifies module
document_name	Module1_Notes.pdf	Identifies source file
page_no	12	Enables source location
chunk_id	M01-P12-C04	Unique evidence unit
text	Extracted passage...	Retrieval content

version	v1.0	Supports document updates

APPENDIX B – MODULE 1 IMPLEMENTATION CHECKLIST

Checklist ID	Module 1 Task	Owner	Evidence to Attach
M1-01	Collect approved course documents	B.Vani	File manifest
M1-02	Validate file types	B.Vani	Validation log
M1-03	Extract text	B.Vani	Extraction sample
M1-04	Clean text	B.Vani	Before/after sample
M1-05	Chunk content	B.Vani	Chunk sample
M1-06	Attach metadata	B.Vani	Metadata table
M1-07	Build index	B.Vani	Index statistics
M1-08	Test retrieval	B.Vani	Retrieval test sheet

Appendix B.1 Retrieval Evaluation Worksheet

Query ID	Expected Topic	Top-1 Relevant?	Top-3 Relevant?	Notes
R01	Course definition	Yes / No	Yes / No	
R02	Module concept	Yes / No	Yes / No	
R03	Process/procedure	Yes / No	Yes / No	
R04	Comparison	Yes / No	Yes / No	
R05	Out-of-scope question	Yes / No	Yes / No	

APPENDIX C – MODULE 2 IMPLEMENTATION CHECKLIST

Checklist ID	Module 2 Task	Owner	Evidence to Attach
M2-01	Prepare grounded prompt	N.R.Yasaswini	Prompt version
M2-02	Assemble retrieved context	N.R.Yasaswini	Context sample
M2-03	Generate answer	N.R.Yasaswini	Output sample
M2-04	Attach source references	N.R.Yasaswini	Citation sample
M2-05	Evaluate relevance	N.R.Yasaswini	Rubric sheet
M2-06	Evaluate faithfulness	N.R.Yasaswini	Claim-evidence sheet
M2-07	Evaluate completeness	N.R.Yasaswini	Expected-points sheet
M2-08	Handle insufficient evidence	N.R.Yasaswini	Abstention sample

Appendix C.1 Claim-to-Evidence Evaluation

Answer Claim	Supporting Source	Supported?	Reviewer Note
Claim 1	Document / page / chunk	Yes / No	
Claim 2	Document / page / chunk	Yes / No	
Claim 3	Document / page / chunk	Yes / No	
Claim 4	Document / page / chunk	Yes / No	

APPENDIX D – USER ACCEPTANCE TESTING

UAT ID	User Task	Expected Behavior	Acceptance
UAT-01	Ask a definition question	Concise grounded answer	Accept / Reject
UAT-02	Ask a concept explanation	Answer with supporting evidence	Accept / Reject
UAT-03	Ask a multi-part question	Addresses available parts	Accept / Reject
UAT-04	Ask outside-course question	System qualifies/abstains	Accept / Reject
UAT-05	Inspect source reference	Source is identifiable	Accept / Reject
UAT-06	Repeat same question	Consistent behavior within configured limits	Accept / Reject

Appendix D.1 User Feedback Form

Question	Rating 1–5	Comment
Answer was relevant		
Answer was understandable		
Answer was sufficiently complete		
Source information was useful		
Response time was acceptable		
Overall satisfaction		

APPENDIX E – PROJECT MANAGEMENT AND QUALITY ASSURANCE

Activity	**Planned Evidence**	**Quality Gate**
Requirements	Signed requirements table	Scope approved
Design	Architecture + workflow figures	Design review
Implementation	Module 1 and Module 2 checklist	Functional review
Testing	Test cases and results	No critical open failures
Security	Threat and control matrix	Security review
Documentation	Chapters, references, appendices	Formatting review
Final submission	Checklist and files	Faculty verification